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Create an ASP.NET Core Web API endpoint that accepts a number and returns its factorial. Test the API using Postman with valid input values.

Project type: ASP.NET Core Web API

Steps to perform

1. In Visual Studio create an ASP.NET Core Web API project.

2. Add Models and Controllers files; configure required services in Program.cs.

3. If EF Core is used, install matching provider packages, set the connection string and apply migrations.

4. Build and run the application.

5. Test GET/POST/PUT/DELETE endpoints using Postman or Swagger where configured.

Closest supplied practical: "AllowedHosts": "*" } Create an API endpoint to calculate Factorial of a Number and test it using Postman.

Source: Dot_Net_Practicals_6_to_10_Organized_v2.pdf

Provided creation instructions and source files

```
Creation steps: Create the FactorialAPI Web API project. Add MathController.cs with the factorial
endpoint
and configure launch settings; run the API and test the endpoint.
Filename: MathController.cs    Location: Controllers/MathController.cs
using Microsoft.AspNetCore.Mvc;
namespace FactorialAPI.Controllers
{
    [Route("api/[controller]")]
    [ApiController]
    public class MathController : ControllerBase
    {
        [HttpGet("factorial/{number}")]
        public IActionResult Factorial(int number)
        {
            if (number < 0)
            {
                return BadRequest(new
                {
                    message = "Factorial is not defined for negative numbers."
                });
            }
            int factorial = 1;
            for (int i = 1; i <= number; i++)
            {
                factorial *= i;
            }
            return Ok(new
            {
                number = number,
                factorial = factorial
            });
        }
    }
}
Filename: LaunchSettings.json    Location: Properties/launchSettings.json
{
  "profiles": {
    "http": {
      "commandName": "Project",
      "environmentVariables": {
        "ASPNETCORE_ENVIRONMENT": "Development"
      },
      "dotnetRunMessages": true,
      "launchBrowser": true,
      "applicationUrl": "http://localhost:5293"
    },
    "https": {
      "commandName": "Project",
      "environmentVariables": {
        "ASPNETCORE_ENVIRONMENT": "Development"
      },
      "dotnetRunMessages": true,
      "applicationUrl": "https://localhost:7103;http://localhost:5293"
```

```
    }
  }
}
```

```
Create a PWA with a Web App Manifest file including app name, icons, start URL and
```

Expected result / verification

The application should meet the exact question requirements. Record actual output or screenshots after running; no execution has been claimed in this manual.